

Store overview

How much do households spend, on what, and when do they shop?

Store overview

Customers

Baskets

Campaigns

Half

H1 (weeks 16-58)

H2 (weeks 59-101)

Week range

16

101

Department

All

Brand

National

Private

Clear filters

Total sales

86 weeks of trading

\$6.8M

Households

93 baskets per household

2.5K

Baskets

2,685 baskets per week

230.9K

Average basket

47% of sales on promotion

\$29.51

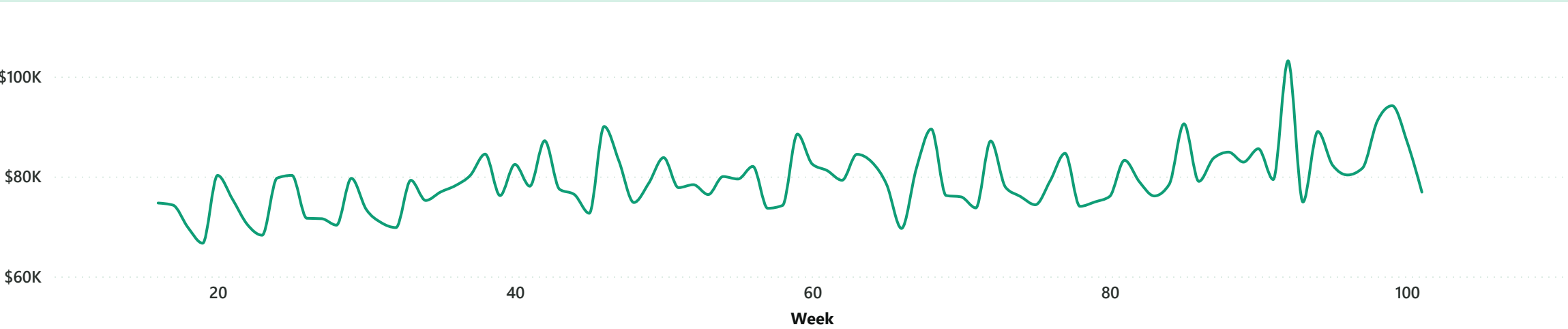
Spend per household per week

Baseline metric the actions should move

\$31.78

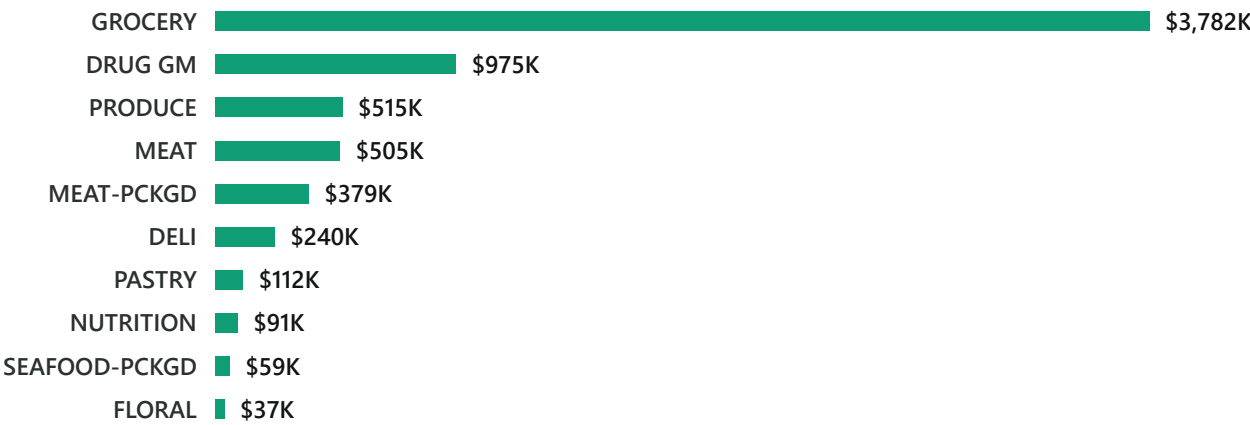
Weekly sales trend

Total sales by week. Hover a week for baskets, households and average basket.



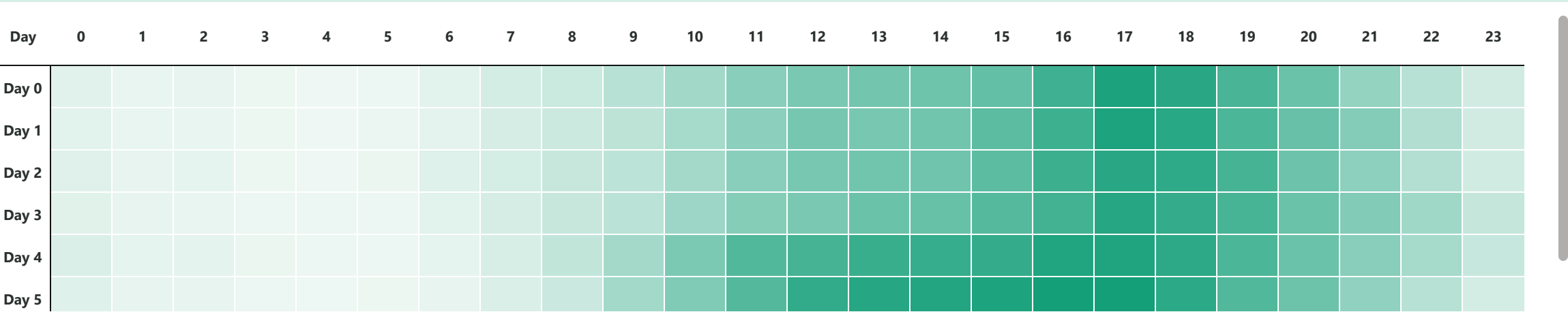
Sales by department

Sorted by sales. Hover for share, promo and private label.



When households shop

Baskets by hour (columns) and relative day (rows). Darker means busier. Hover a cell for the count.



Promotion and private label by department

Top 10 departments by sales: share of each department's sales on promotion and from private label.

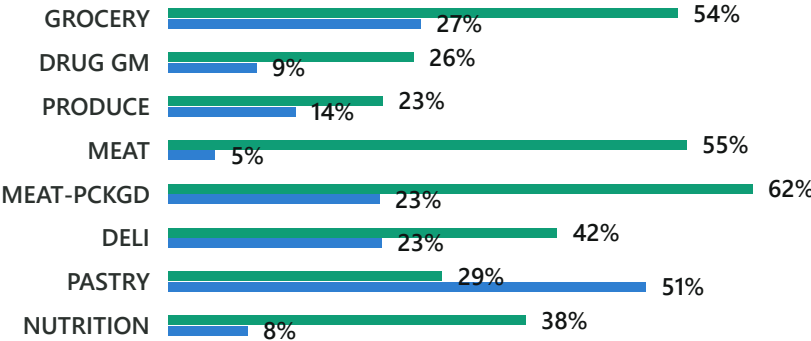
All sales on promotion

All sales private label

Coupon discounts

\$37,208

On promotion Private label



RECOMMENDATION

Use this page as the baseline. Sales per household per week (\$31.78) is the number every action on the next three pages should move.

DATA NOTE

Weeks 16 to 101. Day labels are relative (Day 0 is the first day of data), not real weekdays.

Customers

Who drives sales, who is slipping away, and do lapsed households come back?

Store overview

Customers

Baskets

Campaigns

Half

H1 (weeks 16-58)

H2 (weeks 59-101)

Week range

16

101

RFM segment

All

Frequency tier

All

Income (801 households)

All

Clear filters

Households

594 are Champions

2.5K

Champions share of sales

from 24% of households

50.5%

Lapsed

passed their own lapse threshold

69.7%

Returned after lapse

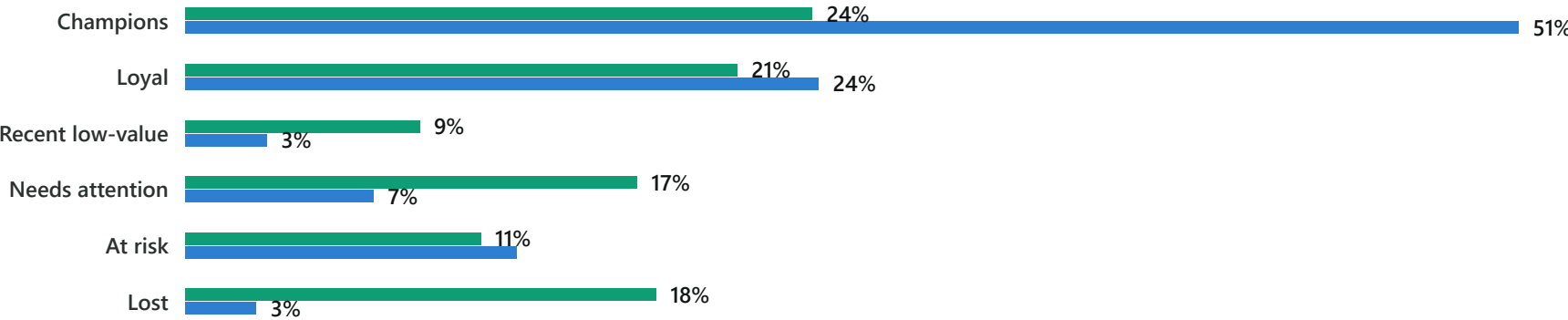
of lapsed households came back

98.2%

Households vs sales by RFM segment

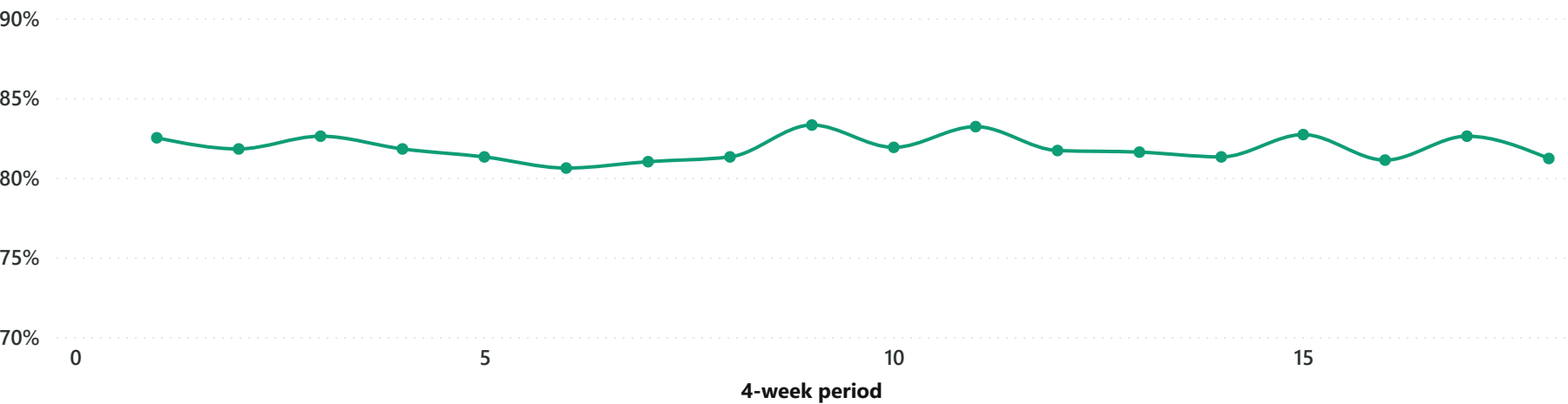
Champions are a quarter of households but half of sales.

● Share of households ● Share of sales



Active households by 4-week period

Share of households that shopped in each period. Axis starts at 70%. Fixed result from notebook 05.



At-risk households to win back

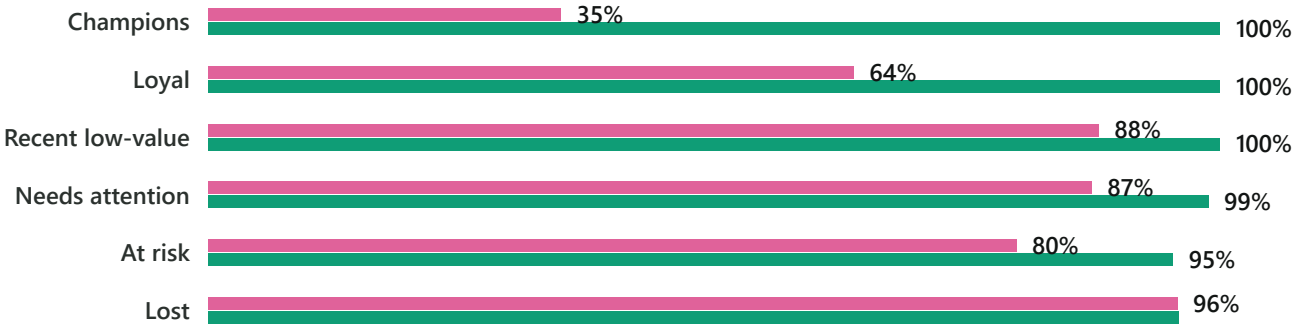
RFM segment 'At risk', highest weekly spend first. Contact them before they pass their own lapse threshold.

Household	Frequency tier	Spend per week	Days since last visit	Lapse threshold (days)	Income	Total sales
232	3: heavy	\$115.90	33	23	35-49K	\$9,968
2318	3: heavy	\$104.12	83	23	Under 15K	\$8,954
149	3: heavy	\$94.56	54	23	35-49K	\$8,132
2294	3: heavy	\$89.12	23	23	25-34K	\$7,575
1110	Not in baseline	\$86.57	15		No demographics	\$5,454
1529	3: heavy	\$85.27	11	23	50-74K	\$7,333
Total						\$854,851

Lapse and return by segment

Share of households that lapsed, and share of those that came back.

● Lapsed ● Returned after lapse



RECOMMENDATION

Protect Champions (24% of households, 51% of sales) and trigger win-back on each household's own purchase rhythm: 98% of lapsing households come back.

DATA NOTE

Income exists for 801 of 2,500 households. Week slicers change sales, not segment counts. Retention line is a fixed result.

Baskets

Which products are bought together, and which pairs cross departments?

Store overview

Customers

Baskets

Campaigns

Pair type

Cross-department

Same department

Department (first item)

All

Lift range

0.70

16.40

Baskets with pair

462

28791

Clear filters

Commodity pairs

10,836,479 pair occurrences in baskets

6.5K

Cross-department pairs

of pairs span two departments

58.9%

Average lift

lift 1.0 means no association

2.47

Strongest pair (lift)

CAT FOOD + CAT LITTER

16.40

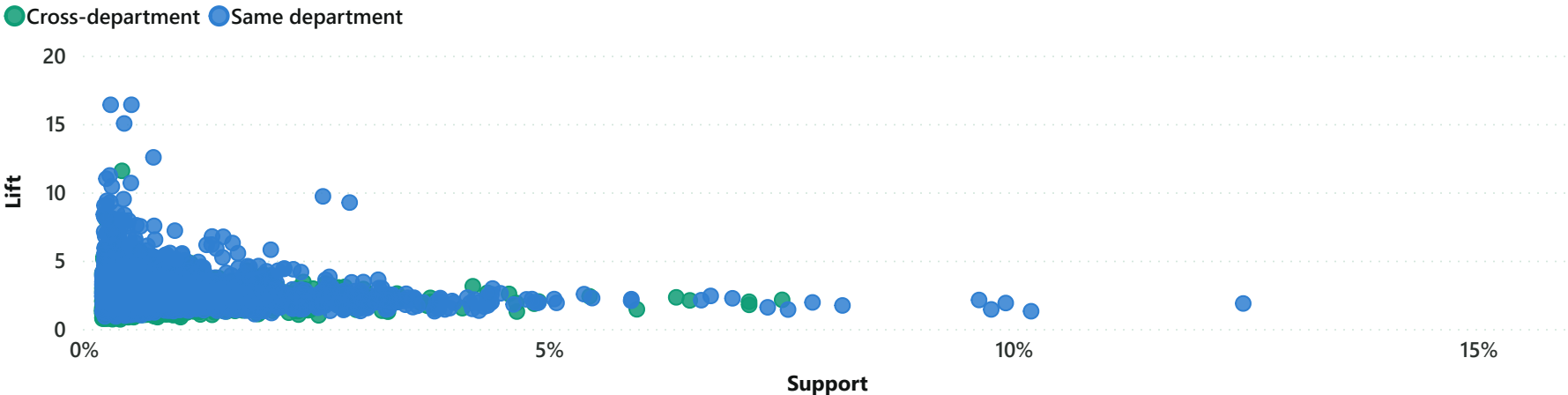
Top pairs by lift

Lift above 1 means the pair is bought together more often than chance.

Pair	Dept A	Dept B	Lift	Support	Confidence A to B	Baskets
CAT FOOD + CAT LITTER	GROCERY	GROCERY	16.40	0.51%	19.4%	1,185
BABY FOODS + INFANT FORMULA	DRUG GM	DRUG GM	16.40	0.29%	17.1%	669
BABY HBC + DIAPERS & DISPOSABLES	DRUG GM	DRUG GM	15.03	0.44%	27.4%	1,005
LAUNDRY ADDITIVES + LAUNDRY DETERGENTS	GROCERY	GROCERY	12.55	0.75%	42.4%	1,731
BROOMS AND MOPS + HOUSEHOLD CLEANG NEEDS	DRUG GM	GROCERY	11.56	0.41%	34.2%	950
BLEACH + LAUNDRY ADDITIVES	GROCERY	GROCERY	11.23	0.28%	19.9%	646
DEODORANTS + SHAVING CARE PRODUCTS	DRUG GM	DRUG GM	10.99	0.24%	14.5%	563
BLEACH + LAUNDRY DETERGENTS	GROCERY	GROCERY	10.67	0.51%	36.0%	1,171
DOG FOODS + PET CARE SUPPLIES	GROCERY	GROCERY	10.42	0.30%	10.5%	697
DRY NOODLES/PASTA + PASTA SAUCE	GROCERY	GROCERY	9.70	2.57%	45.6%	5,940
AIR CARE + HOUSEHOLD CLEANG NEEDS	GROCERY	GROCERY	9.49	0.43%	28.0%	990
HAND/BODY/FACIAL PRODUCTS + SOAP - LIQUID & BAR	DRUG GM	DRUG GM	9.37	0.25%	22.0%	577
BABY FOODS + DIAPERS & DISPOSABLES	DRUG GM	DRUG GM	9.26	0.29%	16.9%	661
CHEESES + DELI MEATS	DELI	DELI	9.24	2.86%	63.0%	6,603
BABY FOODS + BABY HBC	DRUG GM	DRUG GM	9.05	0.24%	14.4%	563
HERBS + MUSHROOMS	PRODUCE	PRODUCE	9.03	0.22%	17.1%	515

Lift vs support

Each dot is a pair. Top right pairs are both common and strong.



Department lift matrix

Average lift of pairs between departments. Darker means stronger association.

Dept A	COSMETI CS	DELI	DRUG GM	GROCERY	MEAT	MEAT-PCKGD	NUTRITIO N	PASTRY	PRODUCE	SALAD BAR	SEAFOOD	SEAFOOD -PCKGD
COSMETICS			4.07	1.63					1.91			
DELI		4.33	1.97	2.14	2.36	2.05	2.17	3.21	2.55	1.30		2.32
DRUG GM	3.97	1.20	4.17	1.71	1.42	1.46	1.66	1.66	1.52	0.92		1.22
GROCERY	1.54	1.89	1.92	2.65	2.50	2.60	2.20	1.70	2.35	0.99	2.08	2.60
MEAT	1.40	1.70	1.49	2.41	3.23	2.67	1.83	1.64	2.64	0.99	2.56	3.09
MEAT-PCKGD	1.41	2.27	1.78	2.76	3.23	3.63	1.39	1.68	2.39	0.92	1.87	3.26
NUTRITION				2.64		2.05			3.09			
PASTRY		2.60	1.15	1.66	1.72	1.56	3.03		2.39			2.21
PRODUCE		2.51	1.97	2.50	2.84	2.23	3.65	2.31	3.96	1.12	3.04	2.92

RECOMMENDATION

Place cross-department complements together (for example brooms next to cleaning supplies) and turn the highest-lift pairs into bundle offers.

DATA NOTE

6,499 commodity pairs from notebook 06. The department slicer filters on the first item of each pair.

Campaigns

Did campaigns really raise household spending, or did they target big spenders?

Store overview

Customers

Baskets

Campaigns

Campaign type

TypeA

TypeB

TypeC

Campaign

All

Coupon redeemed

Not redeemed

Redeemed

Clear filters

Naive spend gap

targeted vs other households, per week

\$31.40

Causal campaign effect

95% CI -\$0.99 to \$3.05, not significant

\$1.03

Explained by selection

of the naive gap is selection, not effect

96.7%

Coupon redemption

889 of 7,208 offers redeemed

12.3%

Naive vs causal estimate

Extra weekly spend per targeted household. Hover a bar for its 95% CI and p-value.

\$31.40

\$1.83

\$5.61

\$1.03

Naive gap

DiD, unmatched

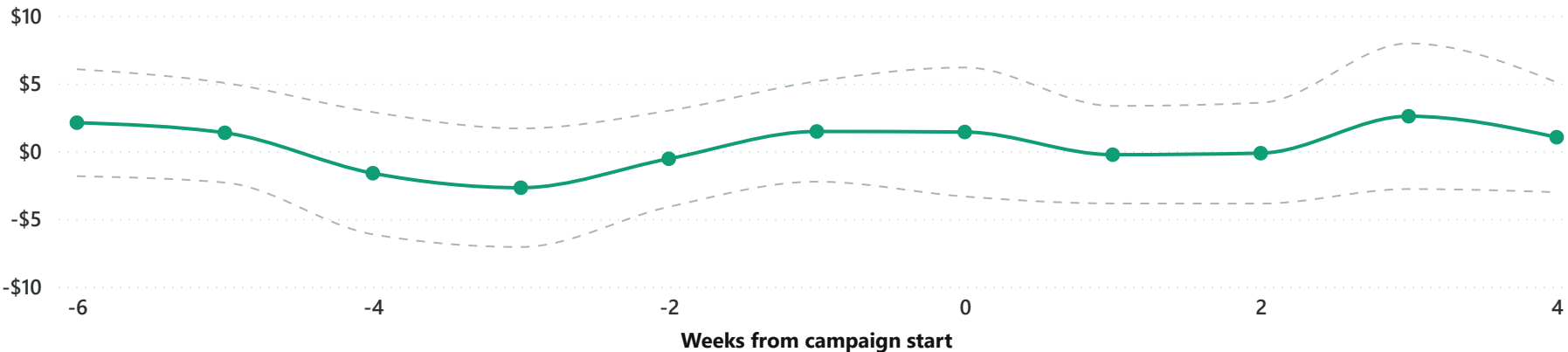
DiD, matched vs never targeted

DiD, matched vs targeted later
(main)

Event study

Weekly effect before and after campaign start (week 0), main control group. The CI crosses zero in every week.

Effect 95% CI low 95% CI high



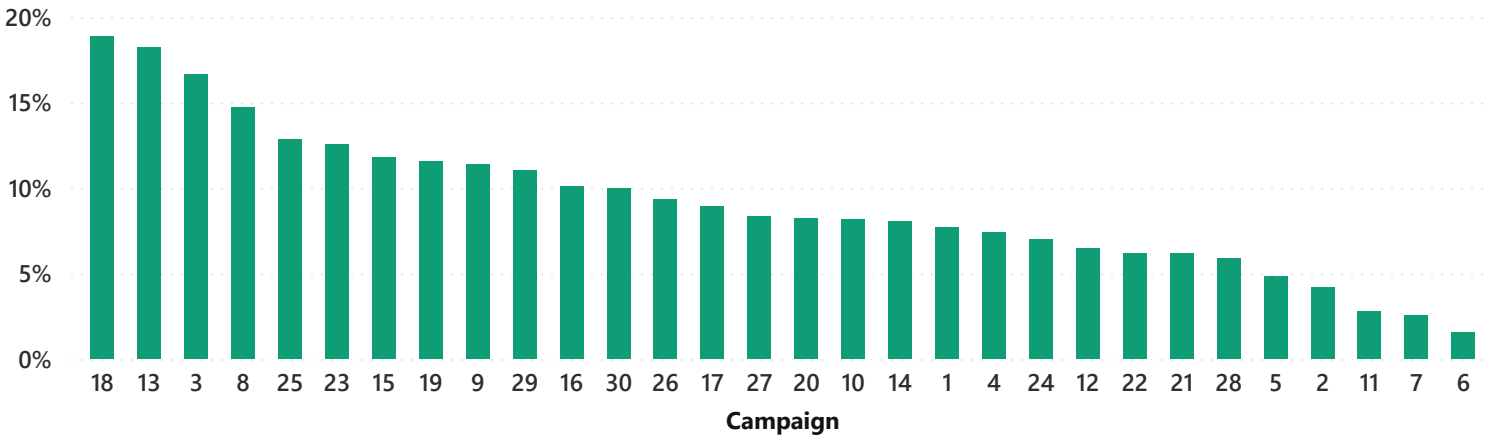
Campaign list

All 30 campaigns. Overlapping weeks make campaign effects hard to separate.

Campaign▲	Type	Start week	End week	Duration (days)	Households targeted	Households redeemed	Redemption
1	TypeB	50	55	38	13	1	7.7%
2	TypeB	51	55	33	48	2	4.2%
3	TypeC	52	60	57	12	2	16.7%
4	TypeB	54	58	33	81	6	7.4%
5	TypeB	55	59	35	166	8	4.8%
6	TypeC	57	61	33	65	1	1.5%
Total					1,584	434	12.3%

Redemption by campaign

Share of targeted households that redeemed a coupon, highest first.



RECOMMENDATION

Add random hold-out groups to every campaign and stop overlapping campaigns: the \$31.40 naive gap shrinks to \$1.03 (not significant) once selection is removed.

DATA NOTE

Effect estimates come from notebook 07 (stacked DiD with propensity matching) and do not change with slicers.